

Question number	Answer	Additional guidance	Mark
3(a)	• 4.2×10^7 (1)		(1)

Question number	Answer	Mark
3(b)	The only correct answer is B <i>A is incorrect because antigenic means can stimulate an immune response</i> <i>C is incorrect because epigenetic means a change in gene expression</i> <i>D is incorrect because polygenic means many genes determining a characteristic</i>	(1)

Question number	Answer	Additional guidance	Mark
3(c)(i)	production of recycled {textiles / clothing} / using waste + that {decreases destruction of habitats / maintains biodiversity / does not harm the environment / reduces production of greenhouse gases / does not increase global warming} (1)		(1)

Question number	Answer
*3(c)(ii)	Positives: • reduces area of land used for cotton farming so {habitats won't be destroyed / biodiversity won't decrease} • fewer vehicles used in cotton production / harvesting / manufacturing decreasing greenhouse gases / reducing global warming • reduces the volume of water used in irrigation so there will be more water available for {humans / wildlife / natural plants} • reduces the mass of textiles being burnt so reduces {greenhouse effect / climate change / greenhouse gases released / toxins} • reduces the use of landfill sites so reduces {habitats destroyed / release of toxins / land for housing issues} • producing recycled clothes provides people with clothing in a more sustainable way • producing recycled clothes provides employment so increases {quality of life / country's economy} • fewer cattle reared for leather for clothing so less methane released • fewer cattle reared for leather for clothing so more land available for growing crops • could reduce the number of items made from crude oil {so decreasing the damage to habitats in mining it / carbon dioxide produced transporting it / conserving its use for other things} Negatives: • new industries need establishing so clothing could be {expensive / unavailable} • fossil fuels still burnt in {machinery used in production / vehicles used in transport} which will still release greenhouse gases • land will still be used in the recycling industries so this will {reduce habitats / decrease biodiversity / cause deforestation} • {waste products will still be produced / dyes will still need removing} which could be toxic

Simple discussion points in context of current methods or recycling:

1 mark = 1 point raised

2 marks = 2 points raised

Extended discussion points in the context of recycling:

3 marks = 2 points raised, plus one which is extended

4 marks = 2 points raised, both of which are extended

3 points raised, one of which is extended

Extended discussion points which illustrate that the sustainable textiles won't completely reduce the conflict:

5 marks = 3 points raised, two of which are extended

4 points raised, one of which is extended